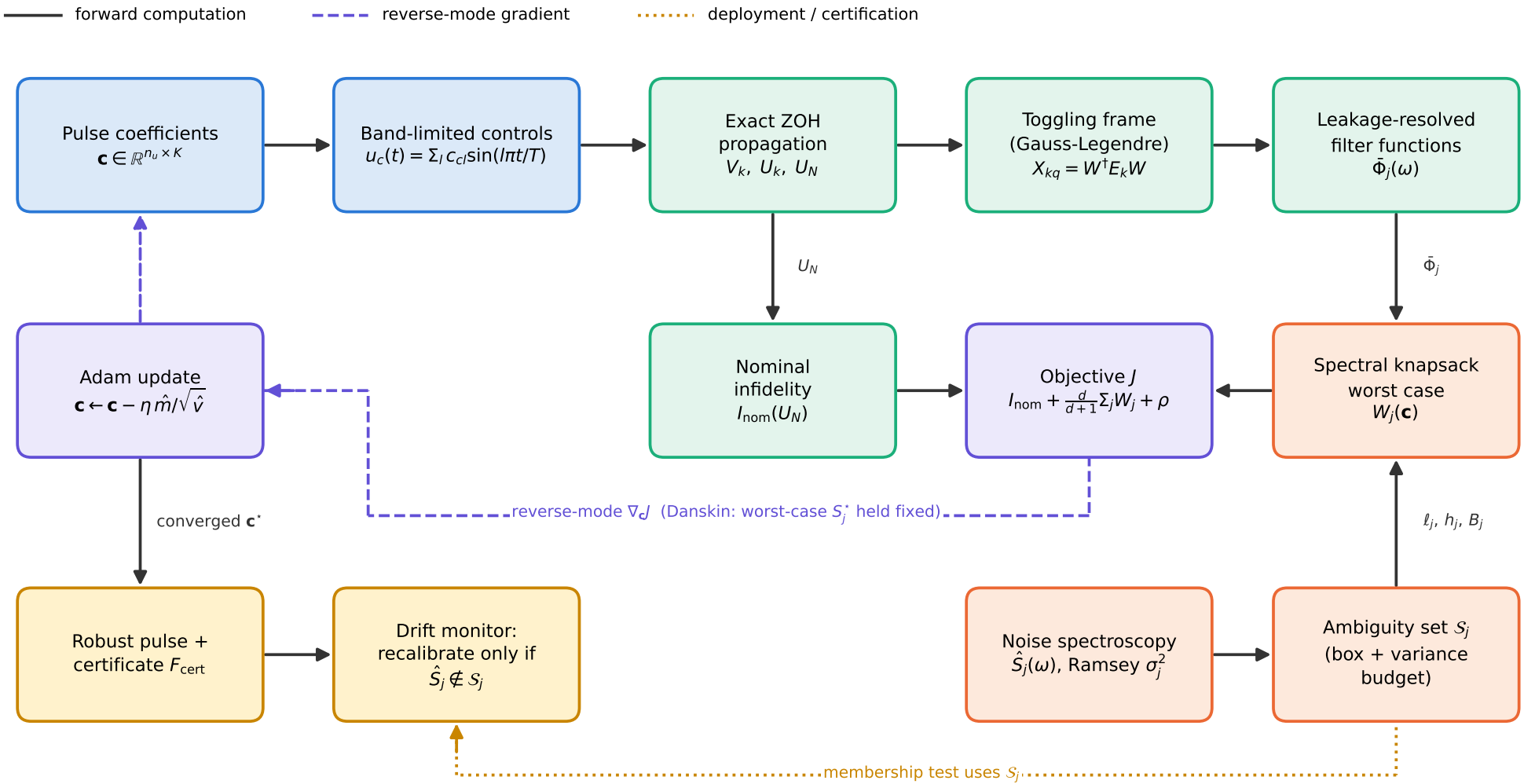
forward computation
reverse-mode gradient
deployment / certification
Pulse coefficients
$\mathbf{c} \in \mathbb{R}^{n_u \times K}$
Band-limited controls
$u_c(t) = \Sigma_l c_{cl} \sin(l\pi t/T)$
Exact ZOH propagation
$V_k,\ U_k,\ U_N$
Toggling frame (Gauss-Legendre)
$X_{kq} = W^\dagger E_k W$
Leakage-resolved filter functions
$\bar{\Phi}_j(\omega)$
U_N
$\bar{\Phi}_j$
Adam update
$\mathbf{c} \leftarrow \mathbf{c} - \eta\, \hat{m}/\sqrt{\hat{v}}$
Nominal infidelity
$I_{\text{nom}}(U_N)$
Objective J
$I_{\text{nom}} + \frac{d}{d+1}\Sigma_j W_j + \rho$
Spectral knapsack worst case
$W_j(\mathbf{c})$
converged $\mathbf{c}^*$
reverse-mode $\nabla_{\mathbf{c}} J$ (Danskin: worst-case S_j^* held fixed)
$\ell_j,\ h_j,\ B_j$
Robust pulse + certificate F_{cert}
Drift monitor: recalibrate only if $\hat{S}_j \notin \mathcal{S}_j$
Noise spectroscopy
$\hat{S}_j(\omega)$, Ramsey σ_j^2
Ambiguity set $\mathcal{S}_j$ (box + variance budget)
membership test uses $\mathcal{S}_j$